

Calc 2 – Parametric Curves

Worksheet A

1. Find $\frac{dy}{dx}$ at $t = 1$ for $x = t^2$, $y = t^3$.
2. Eliminate the parameter in $x = 2 \cos t$, $y = 3 \sin t$, $t \in [0, 2\pi]$.
3. Find the arc length of $x = t$, $y = t^{3/2}$ on $[0, 4]$.
4. Find $\frac{d^2y}{dx^2}$ at $t = 1$ for $x = t^2$, $y = t^3$.
5. Find the area enclosed by the ellipse $x = a \cos t$, $y = b \sin t$ using a parametric area formula.

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Worksheet B

1. Find $\frac{dy}{dx}$ at $t = \pi/4$ for $x = \cos t$, $y = \sin 2t$.
2. Eliminate the parameter in $x = e^t$, $y = e^{2t}$.
3. Find the arc length of $x = \cos t$, $y = \sin t$ on $[0, \pi/2]$.
4. Find the surface area when $x = t$, $y = t^2$, $t \in [0, 1]$ is rotated about the x -axis.
5. Find the area under the cycloid $x = r(t - \sin t)$, $y = r(1 - \cos t)$ for $t \in [0, 2\pi]$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{3t^2}{2t} = \frac{3t}{2}$, at $t = 1$: $\boxed{3/2}$.
2. $\frac{x^2}{4} + \frac{y^2}{9} = \cos^2 t + \sin^2 t = 1$. Ellipse, semi-axes 2 and 3.
3. $x' = 1$, $y' = \frac{3}{2}\sqrt{t}$. $L = \int_0^4 \sqrt{1 + \frac{9t}{4}} dt$. Let $u = 1 + 9t/4$, $du = 9/4 dt$. $L = \frac{4}{9} \int_1^{10} \sqrt{u} du = \frac{4}{9} \cdot \frac{2}{3} (10^{3/2} - 1) = \boxed{\frac{8}{27}(10\sqrt{10} - 1)}$.
4. $\frac{dy}{dx} = \frac{3t}{2} \cdot \frac{d^2y}{dx^2} = \frac{1}{x'(t)} \cdot \frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{1}{2t} \cdot \frac{3}{2} = \frac{3}{4t}$. At $t = 1$: $\boxed{3/4}$.
5. Use $A = \oint y dx = \int_0^{2\pi} b \sin t \cdot (-a \sin t) dt = -ab \int_0^{2\pi} \sin^2 t dt = -ab\pi$. The enclosed area is $|A| = \boxed{\pi ab}$.

Worksheet B.

1. $\frac{dy}{dx} = \frac{2\cos 2t}{-\sin t}$, at $t = \pi/4$: $\frac{2\cos(\pi/2)}{-\sin(\pi/4)} = \frac{0}{-\sqrt{2}/2} = \boxed{0}$.
2. $y = x^2$ for $x > 0$. **Parabola** (right half).
3. $L = \int_0^{\pi/2} \sqrt{\sin^2 t + \cos^2 t} dt = \int_0^{\pi/2} 1 dt = \boxed{\pi/2}$ (quarter unit circle).
4. $S = 2\pi \int_0^1 y \sqrt{(x')^2 + (y')^2} dt = 2\pi \int_0^1 t^2 \sqrt{1 + 4t^2} dt$. Let $u = 1 + 4t^2$, $du = 8t dt$; trig sub or numerical. (Closed form messy.)
5. $A = \int_0^{2\pi} y(t)x'(t) dt = \int_0^{2\pi} r(1 - \cos t) \cdot r(1 - \cos t) dt = r^2 \int_0^{2\pi} (1 - \cos t)^2 dt = r^2 \cdot 3\pi = \boxed{3\pi r^2}$.